"Oh, Give Me a Home (Trade Share)": Differential Import Price Inflation and Gains from Trade Across U.S. Households

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CES 25-47 **July 2025**

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Abstract

Consumers are differentially exposed to trade based on their expenditures, but there is little data on how such trade exposure differs across consumer groups and over time. In this paper, we construct "home trade shares" that vary by age, race, marital status, education, and urban status, and use these to analyze differences in inflation and welfare gains from trade for U.S. demographic groups over the years 1996–2018. We show that over this time period, import prices (inclusive of the effects of taste change) held down overall inflation for all groups. For the typical group, more than a quarter of the gains from trade relative to autarky accrued in our time period. Welfare gains from trade over our time period are largest for rural households, and smallest for Black households. Adding taste change to the typical welfare gains from trade formula boosts the gains for every group relative to the standard formula.

Keyword: import price inflation, gains from trade, inequality

JEL Classification: D12, E31, F14

* Alexis Payne and Brian Green contributed excellent research assistance. We thank Matteo Iacoviello, Hillary Stein, participants at the Syracuse University trade workshop, the USITC Economics seminar, the Federal Reserve International Finance Workshop, the University of Albany Economics seminar, the 2023 System Committee on International Economic Analysis, the 2023 Society for Economic Measurement conference, and the Inter-American Development Bank for helpful suggestions. Any views expressed are those of the authors and not those of the Board of Governors of the Federal Reserve System or any other person associated with the Federal Reserve System. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number #2407 (CBDRB-FY22-P2407-R9809, CBDRB-FY23-P2407-R10276, CBDRB-FY24-0130)
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1 Introduction

How exposed are different types of consumers to international trade? This question matters not only for quantifying the welfare gains to consumers from changes in international trade flows, but especially for understanding the distributional nature of those changes. For example, have white households or black households benefited more from increased U.S. imports over the past three decades? What about urban or rural households? Or older or younger households? Since each of these types of consumers will likely have different consumption baskets, they potentially have a different share of imports in their overall expenditure. Standard results from the international trade literature (e.g., Costinot and Rodríguez-Clare (2014)) directly link the gains from trade to this import share, as well as to the elasticity of international trade to relative prices. However, although it is relatively straightforward to generate an aggregate import share of consumption, there is no comparable data available for different demographic groups in the United States. In this paper, we construct "home trade shares" (the inverse of "import shares") that vary over time and by age, race, marital status, education, and urban status, and together with price data, use these to analyze differences in inflation, the response to import price shocks, and welfare gains from trade for these U.S. demographic groups over the years 1996–2018.

Our starting point is a nested Constant Elasticity of Substitution (CES) preference model in which key parameters are allowed to vary by group. Households purchase both domestic and foreign aggregates, with associated domestic and import price indexes whose components vary across household group, and an overall group-level price index that aggregates these. For the decision of how much of the domestic (foreign) aggregate to buy, demand shifters, price indexes, and price elasticities all can vary by group. The import price index also depends on group-specific sectoral demand shifters. Under this framework, we show that with data on overall expenditures, imported expenditures, overall price indexes, and import price indexes that all vary by group and over time, it is possible to recover the remaining parameters of the model, perform counterfactuals, and estimate the gains from trade.

To construct these data, we rely on microdata from the Consumer Expenditure Survey (CE) from 1996 to 2018. We compute annual expenditures at the detailed product category for household characteristics within 5 demographic categories: Age, Education, Marital Status, Race, and Urban/Rural.[1] Using this expenditure data together with information on the set of consumer-facing imports and import penetration rates for those imported products produces estimates for spending on imports (and home trade shares) for each demographic

[1] In the CE, demographic characteristics are not available for all household members, so these characteristics are for the head of household.

group over the years 1996 through 2018. For price data, we take category-level prices from Bureau of Labor Statistics (BLS) price data underlying the official U.S. Consumer Price Index (CPI) and variety-level prices from restricted access U.S. import data. Combining these prices with their respective expenditure shares allows us to generate overall price indexes and import price indexes that vary by group. These indexes reflect across-category differences across groups but assume away within-category differences across groups given the absence of within-category, group-level data for the universe of consumer goods.

Averaging across groups, home trade shares tend to be around 92% in 1996 and about 89% in 2018. There is variation across groups: comparing 1996 to 2018, rural households and households headed by high school graduates decreased their spending on domestic goods by the most (home trade shares falling by about 4 percentage points), with Black households decreasing their spending on domestic goods by the least (home trade share falling by about 1 percentage point).

Adding on the price data allows us to estimate the elasticity of substitution between domestic and foreign composites (the Armington elasticity) that can vary across groups, as well as group-specific inflation. We find a median Armington elasticity of about 1.5 when estimating group-by-group. For some demographic groups, such as rural households, we find that import prices grew notably slower than overall prices, while for many other groups, import prices grew faster. Even though import prices grew faster than overall prices in many cases, "adjusted" import prices accounting for changes in home trade shares due to taste change indicate that imports put notable downward pressure on overall price inflation over our time period for every group.

Using counterfactual exercises, we estimate that imports held down overall cumulative inflation by about 10 percentage points for the typical group (although the effect was only about 4 percentage points for Black households). In another counterfactual exercise, we find that a uniform 20 percentage point increase in trade costs raises overall prices for young households the most (by 2.5 to 2.2 percent, depending on specification), with a much smaller effect on households with heads over age 60 (by 1.8 to 1.6 percent).

We find sizable heterogeneity in the welfare gains from trade from 1996-2018 across groups. For example, we estimate that rural households were among those with the largest welfare gains from trade over this period (9 to 11.8 percent, depending on specification), while Black households had the lowest gains from trade (3.4 to 4.4 percent). We show how to account for taste change in the otherwise typical welfare gains from trade formula, and find that taste change boosts the gains from trade for every group relative to the standard formula (by about 2/3 ppt. for the typical group, with heterogeneity). As another example, comparing the welfare gains from trade over our observed period to the gains relative to

autarky, we estimate for the typical group that more than one-quarter of the gains relative to autarky accrued in our observed period. However, for Black households only a bit more than 10 percent of the gains relative to autarky occurred over 1996-2018. Furthermore, we split our sample and compute welfare gains from trade over the 1996-2002 and 2002-2018 subperiods, and find interesting heterogeneity across groups. In the early period, we find that some groups such as Black households experienced welfare losses from trade. Finally, we show suggestive evidence that income is not an omitted factor driving our group-level results.

Our paper contributes to a growing literature in international economics on distributional effects of trade via the consumption channel, such as Fajgelbaum and Khandelwal (2016), Borusyak and Jaravel (2017), Carroll and Hur (2020), and Waugh (2023). Our paper differs from these papers in several dimensions: we construct home trade shares over time, we study the welfare gains from trade over an observed time period, we include the effects of taste change, and we study demographic groups other than income groups. None of the prior work does these things. Furthermore, although our demographic group results aren't perfectly comparable to the income group results in these papers, there are some interesting differences nonetheless. For example, our gains from trade results over 1996-2018 don't have a clear pro-low-income bias as would be expected from Fajgelbaum and Khandelwal (2016), nor do our estimated trade elasticities have as clear a pattern across groups as assumed in Waugh (2023).

The rest of the paper is organized as follows. Section 2 lays out the theory and discusses the implementation of the model, while Section 3 presents our results including the differences in the welfare effects of trade across groups. Section 4 concludes.

2 Theory and Implementation

2.1 Framework

We assume U.S. households have nested CES preferences, with the upper-tier a CES aggregate over home and foreign consumption aggregates, such that the utility maximizing expenditure of household group g at time t on home aggregate H ($Y_{H,gt}$), is given by:

$$Y_{H,gt} = \left(\frac{\varphi_{H,gt}^{\sigma_g - 1} P_{H,gt}^{1-\sigma_g}}{P_{gt}^{1-\sigma_g}} \right) Y_{gt}, \tag{1}$$

with elasticity of substitution $\sigma_g > 1$, price of the Home aggregate $P_{H,gt}$, taste shifter $\varphi_{H,gt}$, overall price index P_{gt}, and total expenditure Y_{gt}, all of which can vary by group.

Total expenditure of group g on the foreign aggregate F at time t is likewise:

$$Y_{F,gt} = \left(\frac{\varphi_{F,gt}^{\sigma_g - 1} P_{F,gt}^{1-\sigma_g}}{P_{gt}^{1-\sigma_g}} \right) Y_{gt}, \tag{2}$$

The price index for total consumption of group g is:

$$P_{gt} = \left(\varphi_{H,gt}^{\sigma_g - 1} {P_{H,gt}}^{1-\sigma_g} + \varphi_{F,gt}^{\sigma_g - 1} {P_{F,gt}}^{1-\sigma_g} \right)^{\frac{1}{1-\sigma_g}}. \tag{3}$$

We assume spending on the foreign aggregate is also a CES aggregate over imported sectors, with group-specific taste shifters φ_{gst} that differ by sector, such that the utility maximizing expenditure of group g at time t on imported sector s is given by:

$$Y_{gst} = \left(\frac{\varphi_{gst}^{\sigma_F - 1} {P_{st}}^{1-\sigma_F}}{P_{F,gt}^{1-\sigma_F}} \right) Y_{F,gt}, \tag{4}$$

where P_{st} is the price index of imported sector s, $P_{F,gt}$ is the group-level price aggregate for foreign consumption, $\varphi_{gst} > 0$ is a demand shifter for sector s at time t, and $\sigma_F > 0$ is the elasticity of substitution across imported sectors.

The price index for imported consumption of group g is:

$$P_{F,gt} = \left(\sum_{s \in \overline{S_t}} \varphi_{gst}^{\sigma_F - 1} {P_{st}}^{1-\sigma_F} \right)^{\frac{1}{1-\sigma_F}}, \tag{5}$$

where $\overline{S}_t$ is the set of imported sectors at time t and P_{st} is a sectoral price aggregate which will be defined later.

Lastly, the "home trade share" for group g at time t is defined as $\lambda_{H,gt} \equiv \frac{Y_{H,gt}}{Y_{gt}} = 1 - \frac{Y_{F,gt}}{Y_{gt}}$.

2.2 Welfare

As shown in Arkolakis et al. (2012) and Costinot and Rodríguez-Clare (2014), the gains from trade in this setup can be inferred from how the openness of a country (in their case) is affected, depending on the trade elasticity. In our case, defining real income for group g at any time t as W_{gt}, the standard formula for welfare gains from trade (meaning, the percentage change in real income between two states 0 and 1) is[2]:

$$\dot{W}_{g0} = \ln W_{g1} - \ln W_{g0} = \frac{1}{1-\sigma_g} \left(\ln \lambda_{H,g1} - \ln \lambda_{H,g0} \right) \tag{6}$$

[2]We derive this equation under our framework in Appendix A.

Typically, the welfare gains from trade calculation is done comparing the observed data to autarky, in which $\lambda_{H,g0}$ is set to 1 in Equation (6). We can thus calculate overall welfare gains (relative to autarky) for each group g with data on their home trade share at some time and the trade elasticity. However, it is also possible to calculate welfare gains between any two observed time periods. We have U.S. trade and expenditure data from 1996-2018, so we can also set $\lambda_{H,g1} = \lambda_{H,g2018}$ and $\lambda_{H,g0}$ to be $\lambda_{H,g1996}$. Although in prior work the taste parameters $\{\varphi_{F,gt}, \varphi_{H,gt}\}$ from the preference structure are assumed to be unchanged between time periods, we show in Appendix A that allowing for taste changes between time periods (in the spirit of Redding and Weinstein (2020)) gives rise to a modified version of Equation (6):

$$\dot{W}^{\varphi}_{g0} = \ln W_{g1} - \ln W_{g0} = \frac{1}{1-\sigma_g}\left(\ln \lambda_{H,g1} - \ln \lambda_{H,g0}\right) - \left(\ln \varphi_{Hg1} - \ln \varphi_{Hg0}\right) \tag{7}$$

Thus we also present results from using Equation (7) for 1996 and 2018 data with our estimates of taste changes over that time, which are recoverable from price and expenditure data under our framework. We do not present the results of Equation (7) relative to autarky, since $\{\varphi_{F,gt}, \varphi_{H,gt}\}$ under autarky are not recoverable.

2.3 Estimating Parameters and Recovering Unobservables

We show in this section how, given these data on P_{gt}, $P_{F,gt}$, Y_{gt}, and $Y_{F,gt}$, we can estimate the elasticity of substitution σ_g, recover the taste parameters $\{\varphi_{F,gt}, \varphi_{H,gt}\}$ and generate the (unobservable) domestic price index $P_{H,gt}$.

We start with the estimation of the elasticity of substitution between domestic and foreign varieties. Equation (2) can be re-arranged and written in logs as:

$$\ln\left(\frac{Y_{F,gt}}{Y_{gt}}\right) = (1-\sigma_g)\left[\ln P_{F,gt} - \ln P_{gt}\right] + (\sigma_g - 1)\ln \varphi_{F,gt} \tag{8}$$

Given our time-series data, we will estimate Equation (8) via Ordinary Least Squares for each group g to obtain estimates of the elasticity parameter σ_g, where changes in the taste parameter are the error component [3].

After estimating the elasticity parameter σ_g, we can again rearrange Equation (2) in logs to isolate the taste parameter for foreign purchases, $\varphi_{F,gt}$:

$$\ln \varphi_{F,gt} = \frac{1}{\sigma_g - 1}\ln\left(\frac{Y_{F,gt}}{Y_{gt}}\right) + \left(\ln P_{F.gt} - \ln P_{gt}\right) \tag{9}$$

[3]We tried IV strategies instead of OLS, such as using group-specific sensitivities to the dollar value of the exchange rate as instruments, but the results were little changed from OLS estimates.

This taste parameter which comes from data on prices and, critically, the share of spending on foreign goods.

Next, we assume that the taste parameter for domestic goods is exactly inversely related with the taste parameter for foreign goods:

$$\ln \varphi_{H,gt} = -\ln \varphi_{F,gt} \tag{10}$$

This assumption is sensible because the share of spending on foreign goods that determines φ_F via Equation 9 must be inversely related with the share of spending on foreign goods that determines φ_H. Assuming this particular form also implies that the geometric average of the taste parameters always equals 1, as in Redding and Weinstein (2020).

Lastly, the price index for purely domestically produced goods is not observed, but we can back it out from data on the overall price index P_{gt}, the share of spending on domestic goods, and the taste parameter for domestic goods by re-arranging Equation (1) in logs:

$$\ln P_{H,gt} = \frac{1}{1-\sigma_g} \ln \left(\frac{Y_{H,gt}}{Y_{gt}} \right) + \ln \varphi_{H,gt} + \ln P_{gt} \tag{11}$$

2.4 Constructing Expenditures and Price Indexes

In this section, we discuss how we construct group-level total expenditures (Y_{gt}), group-level expenditures on foreign goods ($Y_{F,gt}$), the group-level overall price index (P_{gt}), and the group-level foreign price index ($P_{F,gt}$) from the data.

2.4.1 Group-Specific Expenditure Data

The Consumer Expenditure Survey (CE), maintained by the U.S. Bureau of Labor Statistics (BLS), contains household-level data on expenditures, income, and demographics in the United States, and are the data that underlie the expenditure weights in the U.S. CPI. For our analysis, we use the Public Use Microdata (PUMD), which provides disaggregated expenditure information for individual households (excluding information that could identify them) from 1996-2018. The PUMD also includes adjustments for information that is missing because respondents were unwilling or unable to provide it. Surveyed respondents provide demographic information for the head of household as well as expenditure for various Universal Classification Code (UCC) products, of which there are a total of around 650-670, depending on the year.[4]

[4]These reflect only those categories listed under "Food" or "Expenditure" in the CE hierarchical grouping file, not items listed as "Assets", gifts, or anything else. See `https://www.bls.gov/cex/pumd/stubs.zip` for the full list.

In what follows, mirroring the language of the BLS, we will refer to consumer *characteristics* as individual survey responses within a particular *demographic*. For example, for the demographic "Race of Member", the available characteristics in 2004 are White, Black, Native American, Asian, Pacific Islander, and Multi-Race.[5]

In this paper, we focus on 5 demographics in the CE: Age, Education, Marital Status, Race, and Urban Status. Within those demographics, we analyze 13 household characteristic groups: Under Age 30, Age 30-60, Age 60 and over, High School Graduate, College Graduate, Post-Graduate, Unmarried, Married, White, Black, Asian and Pacific Islander, Urban, and Rural.[6] Household characteristics within a demographic will not overlap. Note that these are not the only characteristics available for these demographics. However, since the PUMD is a survey, the data becomes less representative of the overall population if there are very few respondents.[7] We left out certain characteristics and combined certain other characteristics together for this reason.[8] We additionally prioritized having consistently-defined characteristics over the entire time frame of our analysis.[9] Finally, we note that these characteristics are self-reported: even though about 20% of the U.S. population lives in a rural area according to the official Census Bureau definition, much greater than the share of households in the CE, respondents answer these survey questions according to their own understanding.[10]

For each year t and group g in the CE sample, we can obtain total expenditure for each category j, and can sum these to obtain total household expenditure Y_{gt} for each of the 13 groups. The CE does not, however, separately identify domestic and imported expenditure. To generate $Y_{F,gt}$, we take the category j-level expenditure shares from the CE, and, using the concordance from Furman et al. (2017), convert the expenditures to the Harmonized System (HS) 6-digit nomenclature used for trade data (k). Making a proportionality assumption, we then apply national import penetration ratios for each year t to expenditure to each

[5]The Dictionary for Interview and Diary Surveys has a complete list of all demographics in the CE, and is available at `https://www.bls.gov/cex/pumd/ce_pumd_interview_diary_dictionary.xlsx`. Note that these demographics refer to that of the head of a CE "consumer unit", namely the person whose name appears on the rental contract or deed. We use the term "head of household" throughout the paper to capture this idea.

[6]We use the terms "Black/White/Asian households" as shorthand for households headed by a consumer of race Black/White/Asian; of course, other members of the household could be of a different race.

[7]The BLS includes calibration weights in its CE Public Use Microdata for each household in order to be representative of the full CE sample.

[8]The number of respondents underlying each characteristic we use is summarized in Table D.3.

[9]For example, a variable identifying whether a head-of-household is of Hispanic origin is present in the CE, but was not defined for the entire length of our sample.

[10]See "What is Rural America": `https://www.census.gov/library/stories/2017/08/rural-america.html`.

HS6 category, constructed using data on U.S. imports, exports and production data at the annual level, as in Hottman and Monarch (2020), giving group-specific expenditure on each imported category k.[11] Summing these expenditures up generates $Y_{F,gt}$, which differs across groups as a result of the different underlying consumption basket of each group. In other words, given our assumption that import penetration rates at the product level do not vary across groups, our home trade shares and import price indexes will reflect across-category differences but not within-category differences across demographic groups[12].

Even without bringing on price data yet, the results of this calculation we just discussed already provides a useful object for our analysis, namely, the share of expenditure on domestic goods by group (which is $1 - \frac{Y_{F,gt}}{Y_{gt}}$). This "home trade share" will be a key object in our consideration of the group-level welfare gains from trade, as we showed in Section 2.2.

2.4.2 Group-Specific Aggregate Price Indexes

Generating a price index at a given time period requires, at minimum, two inputs: 1) the share of spending on every category in overall expenditure and 2) the prices of those individual categories. An aggregate price index, such as the CPI, requires aggregate expenditure data and aggregate prices, while group-specific price indexes would require either group-specific expenditure, group-specific prices, or both. Since group-specific category prices are not available in our data, we combine group-specific expenditure data at the category level with aggregate price data for those same categories to generate group-specific price indexes.

In order to generate group-specific aggregate price indexes, we combine the category j-level expenditure shares discussed above with price data that underlies the CPI, also available from the BLS. Specifically, we download all non-seasonally adjusted CPI-category level prices available for all months from January 1996 through December 2018, and collapse the data to quarterly. We then use a concordance from Orchard (2025) between CPI categories and CE UCC categories to have both expenditure shares (which are available at annual frequency) and prices (which are available at quarterly frequency) for all possible UCC codes[13]. Group-specific inflation and price indexes P_{gt} for every quarter 1996:Q1 through 2018:Q4 are then constructed according to:

[11] The import penetration ratio for imported product k is $\frac{M_{kt}}{P_{kt}+X_{kt}-M_{kt}}$, where M_{kt} and X_{kt} are U.S. imports and exports of product k and P_{kt} is production of product k from the BEA. More detail is presented in Appendix B.

[12] We could incorporate within-category differences in imports across groups, but no such data is available for the universe of consumer goods.

[13] This concordance is available as "CEX_CPI_crosswalk.xlsx" from `https://github.com/JakeOrchard/CDS-Public/tree/main/raw_data`.

$$\pi_{gt} = \frac{P_{g,t}}{P_{g,t-1}} = \sum_{j \in \{G_t \cup G_{t-1}\}} \frac{Y_{gjt}}{Y_{gt}} \frac{P_{j,t}}{P_{j,t-1}} \tag{12}$$

where G_t is the set of consumer categories available at time t, Y_{gjt}/Y_{gt} is the group-specific expenditure share on category j at time t, and $P_{j,t}$ is the price for cateogry j from the BLS. The growth in the price index for group g at time t is the weighted average of price index growth across all categories purchased in both t and $t-1$. Note that price index for any given time $P_{g,t}$ can be construced by cumulating up the changes. This is similar to the approach to constructing group-specific price indexes in Jaravel (2024). While (12) is not the same structure as (3), we assume that our empirical price index from the BLS data is a reasonable approximation to the theoretical CES price index.

2.4.3 Group-Specific Import Price Indexes

As previously outlined, in order to construct import price indexes, we use data on prices, sales, and expenditures with a nested CES demand system at the HS6 sectoral level.

Starting from Equation (5), we generate group level chained import price indexes, with the time t link given by:[14]

$$\pi_{F,gt} = \frac{P_{F,gt}}{P_{F,gt-1}} = \frac{\left(\sum_{s \in \overline{S_t}} \varphi_{gst}^{\sigma_F - 1} {P_{st}}^{1-\sigma_F}\right)^{\frac{1}{1-\sigma_F}}}{\left(\sum_{s \in \overline{S_t}} \varphi_{gst-1}^{\sigma_F - 1} {P_{st-1}}^{1-\sigma_F}\right)^{\frac{1}{1-\sigma_F}}}. \tag{13}$$

Unlike Equation (12), the components of Equation (13) are not directly taken from the data, so additional estimation steps are needed. To construct the import price index for each group g in each period (P_{gt}^F) as in Equation (13), we need an estimate of σ_F, sector level price indexes P_{st}, and the taste shifter φ_{gst}. To obtain these, we thus assume that sectoral prices are also a CES aggregate of imported variety-level prices:

$$P_{st} = \left(\sum_{v \in \overline{G}_{st}} {\varphi_{vt}}^{\sigma^s - 1} p_{vt}^{1-\sigma^s} \right)^{\frac{1}{1-\sigma^s}}, \tag{14}$$

where $\overline{G}_{st}$ is the set of varieties v in imported sector s at time t, φ_{vt} is a taste shifter, and σ^s is the elasticity of substitution for s, and p_{vt} is the price of an imported variety. The price of imported sector s is assumed to be identical across groups.

[14] We also define the set of tradable consumer goods sectors at time t to be S_t, with $\overline{S}_t = \{S_{t+1} \cap S_t \cap S_{t-1}\} \cup \{S_t \cap S_{t-1} \cap S_{t-2}\}$ the set of sectors at time t that are a) also present in $t-1$ and b) present for 3 consecutive quarters.

With quarterly prices and sales data at the foreign supplier-HS10 product level from 1996 through 2018 in the Linked-Longitudinal Firm Trade Transaction Database (LFTTD), we apply the Feenstra (1994) estimation strategy to recover σ_F and the other parameters in (14), form P_{st}, and retrieve estimates of φ_{gst}. The estimation is similar to that used in Hottman and Monarch (2020), and is detailed in the Appendix.

This permits the construction of $P_{F,gt}$ via Equation (13). Importantly, when considering the role of import prices they enter into overall inflation as in Equation (3), it is necessary to consider what we call "adjusted" import price inflation that accounts for changes in import prices together with changes in $\varphi_{F,gt}$:

$$\widetilde{\pi}_{F,gt} = \frac{P_{F,gt}}{\varphi_{F,gt}} / \frac{P_{F,gt-1}}{\varphi_{F,gt-1}} \tag{15}$$

3 Results

3.1 Home Trade Shares

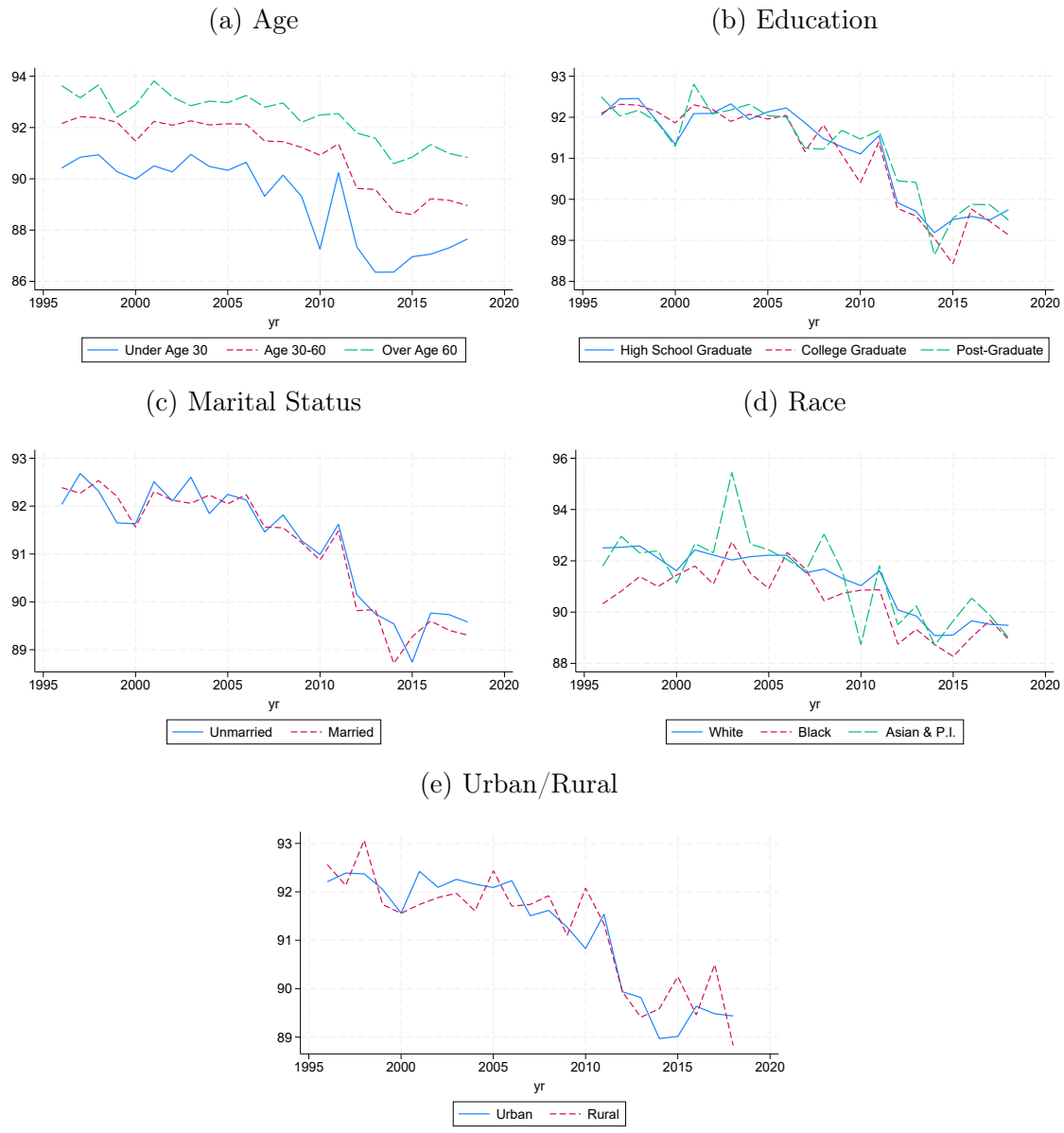
The first 3 columns of Table 1 shows the home trade share for each characteristic in 1996, 2002, and 2018, while Figure 1 shows the time series. Differences across groups are most pronounced for households headed by consumers of different age groups – those with a head under age 30 spend persistently about 3 percentage points less on domestically-produced goods than those with a head over age 60. There are other notable differences: Black households had a more than 2 percentage point lower home trade share in 1996 than White households, while in 2018, households with a head under age 30 had the lowest home trade share.

Another message from Table 1 and Figure 1 is that, consistent with Hottman and Monarch (2020), spending on domestic goods decreased steadily over this time period across all characteristics. As shown in the last column of Table 1, for almost every demographic group, home trade shares are about 2 to 4 percentage points lower in 2018 than in 1996. Black households had the smallest decrease (1.4 percentage points) while rural households had the largest decrease (3.8 percentage points). Most of the changes in home trade shares occurred after 2002, potentially related to developments such as China's entry into the WTO in 2001.

3.2 Inflation by Demographic Group

We next summarize inflation for each group g, π_{gt} and $\pi_{F,gt}$, as described in Equations (12) and (13). Recall that to construct π_{gt}, we combine BLS data on prices of CPI categories

Figure 1: Home Trade Shares (%)



Notes: Home trade shares are calculated by combining national import penetration ratios with group-level expenditure on each HS6 category.

with the group-specific expenditure shares constructed from the CE, while $\pi_{F,gt}$ is constructed from the variety-level import data from the U.S. Census. In both cases, we generate a chained index by constructing the weighted average of prices for the set of goods found in each base year and that same set of goods in the previous year. The set of purchased goods can and does change over time.

The left part of Table 2 provides the geometric average annual inflation rates from 1996–

Table 1: Home Trade Shares (%)

Demographic	Characteristic	1996	2002	2018	Δ Home Trade Share 1996-2018
Age	Under Age 30	90.4	90.3	87.7	-2.7
	Age 30-60	92.2	92.1	89.0	-3.2
	Over Age 60	93.6	93.2	90.8	-2.8
Education	High School Graduate	92.0	92.1	89.7	-2.3
	College Graduate	92.1	92.2	89.1	-3.0
	Post-Graduate	92.5	92.1	89.5	-3.0
Marital Status	Unmarried	92.0	92.1	89.6	-2.4
	Married	92.4	92.1	89.3	-3.0
Race	White	92.5	92.2	89.5	-3.0
	Black	90.3	91.1	88.9	-1.4
	Asian & P.I.	91.8	92.3	89.0	-2.8
Urban/Rural	Urban	92.2	92.1	89.4	-2.8
	Rural	92.6	91.9	88.8	-3.8
	Median	92.2	92.1	89.3	-2.8

Notes: Home trade shares are calculated by combining national import penetration ratios with group-level expenditure on each HS6 category.

2018, for both overall prices and import prices. For some demographic groups (Over Age 60, H.S. Graduate, Unmarried, White, and Rural Consumers), import prices grew slower than overall prices, while for other groups, import prices grew faster. Taking the median across groups, overall prices grew about 2.13% per year, while import prices grew about 2.11% per year. Overall price inflation ranges from 2.07% per year to 2.20%, while import price inflation ranges from 1.34% (Rural) to 2.70% (Black). White households had 0.75 ppt lower annual import price inflation compared to Black households, and Rural consumers had 0.78 ppt lower import price inflation than Urban households per year.[15]

Even though import prices grew faster than overall prices in some cases, the relevant object for considering the role of import prices on overall price inflation is an "adjusted" version of import price inflation that accounts for taste changes. In other words, recalling

[15]Figures E.1 and E.2 show the time series of the price indexes. Figure E.3 plots a BLS-based national consumer import price index based on four broad end-use categories against the import price index for households with a head over 60, and shows that the BLS-based index exhibits qualitatively similar patterns as our group import price indexes.

Table 2: Average Annual Inflation, 1996–2018

Demographic	Characteristic	π_{gt}	$\pi_{F,gt}$	Variable σ_g σ_g	Variable σ_g $\widetilde{\pi}_{F,gt}$	$\sigma_g = 1.46$ $\widetilde{\pi}_{F,gt}$
Age	Under Age 30	2.14	2.34	1.35	-1.09	-0.31
	Age 30-60	2.10	2.12	1.55	-0.67	-1.19
	Over Age 60	2.17	1.68	1.46	-1.31	-1.32
Education	H.S. Grad.	2.14	1.73	1.54	0.05	-0.31
	College Grad.	2.07	2.19	1.48	-0.86	-0.99
	Post-Grad.	2.08	2.11	1.25	-3.76	-1.15
Marital Status	Unmarried	2.20	1.98	1.39	-0.86	-0.39
	Married	2.07	2.18	1.50	-0.90	-1.19
Race	White	2.12	1.95	1.46	-1.10	-1.13
	Black	2.15	2.70	1.40	0.65	0.85
	Asian & P.I.	2.19	2.06	1.47	-0.57	-0.64
Urban/Rural	Urban	2.12	2.12	1.46	-0.81	-0.81
	Rural	2.13	1.34	1.38	-2.61	-1.77
	Median	2.13	2.11	1.46	-0.86	-0.99

Notes: This table shows estimated average annual inflation, import price inflation, and "adjusted" import price inflation for each of the 13 demographic characteristics, implementing Equations (12), (13), and (15). Group-specific elasticity parameters σ_g are estimated by implementing Equation (8).

Equation (3) and (15), it is $\frac{P_F}{\varphi_F}$ that directly feeds into overall prices. To compute φ_F according to Equation (9), estimates of σ_g are required, which come from estimating Equation (8) using home trade shares and prices over the 2000-2018 period.

The right section of Table 2 reports the relevant objects. The third column shows our estimates of σ_g: the median across groups is 1.46, with a high of 1.55 (households with a head age 30-60) and a low of 1.25 (households with a post-graduate head). For comparison, our estimate[16] of σ_F is 1.78, and we estimate a mean σ^s of 4.2 (with a standard deviation of σ^s estimates across sectors of 2.8). The σ_g estimates are fairly low by the standards of the trade literature, although not by the standards of the international macro literature. Our σ^s estimates are comparable to those from Broda and Weinstein (2006). As in Feenstra et al. (2018), σ_g and σ^s are distinct parameters, and thus even with similar σ^s estimates to Broda and Weinstein (2006) we will estimate much larger gains from trade than is typical in the trade literature.

[16]Estimation details are in Appendix C.

The fourth and fifth columns of Table 2 show the average annual growth rate of adjusted import prices $\widetilde{\pi}_{F,gt} = \frac{P_{F,gt}}{\varphi_{F,gt}} / \frac{P_{F,gt-1}}{\varphi_{F,gt-1}}$. The fourth column uses the variable σ_g estimates, while the fifth column reports the same object but with σ_g fixed at the median estimate, 1.46, for all groups. Here, comparing Columns 4 and 5 against Column 1, it can be clearly seen that the adjusted version of import prices always grow slower than overall inflation, and for many groups, outright declined. This finding implies that import prices put downward pressure on overall price inflation over this time period for all groups, but differentially. Imported goods had the largest drag on overall inflation for rural households and households with a head over age 60, while they put the least downward pressure on overall inflation for Black households. We will be more precise about the quantitative effect of imports on inflation using a counterfactual exercise a bit later in the paper.

3.3 Welfare Results and Counterfactual Experiments

We now use the model to consider the effects of a uniform increase in import prices of 20% (implictly coming from a uniform increase in trade costs). Even though the price increase is uniform, because each group is differentially exposed to imports, the effects can be heterogeneous across groups. One simple way to predict the effects of a rise in trade costs on overall prices would be to simply multiply the increase in prices by the share of imported consumption (i.e., one minus the home trade share) for each group (as in Borusyak and Jaravel (2017)). This simple calculation is shown in the first column of Table 3: the 20% increase in import prices would be expected to increase overall prices by 2.14 percent for the median demographic group.

However, our model allows households to substitute away from foreign goods in response to this price shock. To run this experiment using our model, we again start with Equation (3), but now, starting from the 2018Q4 index value, we increase the foreign price index component by 20%. The rightmost 2 columns of Table 3 shows the increases in the overall price index that arise from this change for both variable and constant σ_g terms. As expected, we find a smaller increase in the overall price index arising from this shock to trade prices for all groups, whether using a fixing σ_g or allowing it to vary (with a median increase of about 1.9 percent). However, even though the magnitude differs, since the change in overall prices is very dependent on the home trade share, the ordering of groups is not changed by using any of the three methods: households with a head under age 30 are always most affected, while households with a head over age 60 are always least affected. Our results do show that even with a uniform increase in foreign prices, the effects on overall price indexes do differ across groups.

Table 3: Effects of a 20% Increase in Import Prices in 2018

		Change in Overall Price Index (%)		
Demographic	Characteristic	Simple	Variable σ_g	$\sigma_g = 1.46$
Age	Under Age 30	2.47	2.21	2.19
	Age 30-60	2.21	1.95	1.96
	Over Age 60	1.83	1.62	1.62
Education	H.S. Grad.	2.05	1.80	1.82
	College Grad.	2.17	1.92	1.93
	Post-Grad.	2.10	1.89	1.86
Marital Status	Unmarried	2.08	1.86	1.85
	Married	2.14	1.89	1.90
Race	White	2.10	1.86	1.86
	Black	2.21	1.97	1.96
	Asian & P.I.	2.20	1.95	1.95
Urban/Rural	Urban	2.11	1.87	1.87
	Rural	2.23	2.00	1.98
	Median	2.14	1.89	1.90

Notes: This table shows the counterfactual effect of a 20% increase in import prices in 2018 on the overall price index in 2018, using only home trade shares (column 1) and using our full framework including both variable and fixed σ_g (columns 2 and 3). Source: LFTTD, BLS Consumer Expenditure Survey and authors' calculations.

We next compute welfare gains from trade for each of the different demographic groups. To do so, we use the standard Arkolakis et al. (2012) formula from Equation (6) for 2018 relative to autarky ($\dot{W}_A$) and relative to 1996 ($\dot{W}_{1996}$), as well as our taste-change version of the formula in Equation (7) for 2018 relative to 1996 ($\dot{W}^{\varphi}_{1996}$). We use the home trade shares described above, and our estimates of σ_g. Recall that home trade shares fell from about 92% in 1996 to 89% in 2018 on average and we estimated an elasticity of substitution that had a median value of about 1.46. The left side of Table 4 reports the welfare numbers when we allow σ_g to vary across groups, while the right side reports results using the median estimate for σ_g for each group.

The standard welfare gains calculation comparing 2018 to autarky (a home trade share of 1) is shown in Column 1. The welfare calculations show that the groups benefiting most from imports were households with a head with a post-graduate degree, households with a head under 30, and rural households, mainly groups that had the lowest home trade shares in 2018. These estimates are large relative to other work on static gains from trade, because

Table 4: Welfare Gains from Trade in 2018 (%)

		Variable σ_g			$\sigma_g = 1.46$		
Demographic	Characteristic	$\dot{W}_A$	$\dot{W}_{1996}$	$\dot{W}^{\varphi}_{1996}$	$\dot{W}_A$	$\dot{W}_{1996}$	$\dot{W}^{\varphi}_{1996}$
Age	Under Age 30	37.9	9.0	9.7	28.6	6.8	7.4
	Age 30-60	21.4	6.5	7.1	25.4	7.7	8.4
	Over Age 60	20.8	6.6	7.3	20.9	6.6	7.3
Education	H.S. Grad.	20.0	4.7	5.1	23.5	5.5	6.0
	College Grad.	23.9	6.8	7.5	25.0	7.1	7.8
	Post-Grad.	44.2	13.1	14.5	24.1	7.2	7.9
Marital Status	Unmarried	28.3	7.0	7.6	23.9	5.9	6.4
	Married	22.4	6.7	7.4	24.6	7.4	8.1
Race	White	23.9	7.1	7.8	24.2	7.2	7.9
	Black	29.5	3.9	4.4	25.5	3.4	3.8
	Asian & P.I.	24.7	6.6	7.2	25.3	6.7	7.3
Urban/Rural	Urban	24.2	6.6	7.3	24.3	6.6	7.3
	Rural	31.5	10.9	11.8	25.8	9.0	9.7
	Median	24.2	6.7	7.4	24.6	6.8	7.4

Notes: This table shows the welfare gains from trade in 2018 under differing assumptions on σ_g, calculated using Equation (6) relative to autarky (columns 1 and 4), Equation (6) relative to 1996 (columns 2 and 5), and Equation (7) relative to 1996 (columns 3 and 6).

our elasticity of substitution is low relative to that used in the trade literature.

Over the time period of our sample (columns 2-3), rural households, and households with a head with a post-grad degree, or under age 30 had the highest welfare gains from trade. As can be seen from Table 1, these are the groups that had the largest reductions in their home trade shares over this time period, about 3-4 percentage points. Households with a head who was Black benefited the least. Table 4 also shows that for the typical group more than one-quarter of the overall welfare gains from trade relative to autarky occurred due to changes in home trade shares during our sample period, though for Black households, only 13% occurred during the period. Accounting for taste changes that occur over our time period boosts the welfare gains by about 0.6 ppt. for the typical household, with some sizable differences across households. The results are not appreciably different when fixing σ_g across groups at its median or allowing σ_g to vary, so for the remaining results, we simply fix $\sigma_g = 1.46$ for all groups.

Next, we quantify the effects of imports on overall price inflation across groups. For this

Table 5: Effects of Import Prices Growing Identically to Domestic Prices, 1996-2018

		Cumulative Overall Inflation (%)		
Demographic	Characteristic	Data	$\Delta P_{F,gt} = \Delta P_{H,gt}$	π Gap
Age	Under Age 30	61.8	70.7	8.8
	Age 30-60	60.3	70.9	10.6
	Over Age 60	63.2	73.0	9.8
Education	H.S. Grad.	61.8	69.7	7.9
	College Grad.	59.3	68.9	9.6
	Post-Grad.	59.8	69.7	9.9
Marital Status	Unmarried	64.1	72.6	8.4
	Married	59.4	69.5	10.0
Race	White	61.1	71.3	10.2
	Black	62.3	66.3	4.0
	Asian & P.I.	63.7	73.2	9.5
Urban/Rural	Urban	61.0	70.2	9.2
	Rural	61.4	74.6	13.2
	Median	61.4	70.7	9.6

Notes: The table shows cumulative inflation by demographic characteristic in the data (column 1) as well as under the counterfactual in which import prices are assumed to grow at exactly the same rate as domestic prices did according to our estimates (column 2). Column 3 is the difference between Columns 1 and 2.

experiment, we start with Equation (3) and assume that foreign prices evolved exactly in the same way as domestic goods. Specifically, for each quarter from 1996Q1 through 2018Q4, the growth rate of foreign prices is the same as the growth rate of domestic prices, but the taste parameters $\{\varphi_{H,gt}, \varphi_{F,gt}\}$ (which, as shown in Equations (9) and (10) are determined by the home trade share, σ_g, and the original price indexes) are unchanged from our baseline. In other words, we assume that consumers' tastes for foreign goods are unchanged from how they evolve in the baseline, but foreign prices grow as fast as domestic goods. Recall that although domestic prices are not observable directly, we recover them via Equation (11). Table 5 compares the cumulative increase in the overall price index in our baseline (the data) to the cumulative increase in the overall price index under this counterfactual. When import prices grow as fast as domestic prices, cumulative inflation over our 22 year period is almost 10 ppt. higher for the median demographic group, with cumulative inflation rising from 61% in the baseline to 71% in the counterfactual. Put differently, imports held down overall price inflation by about 10 percentage points for the median group. These

counterfactual "π Gaps" are shown in the last column. Mirroring the results in Table 2, the counterfactual increase in overall inflation is especially pronounced for rural households (13%) and and white households (10.2%), while Black households have the smallest increase (4%). This shows that import prices held down overall price inflation the least for black households and the most for rural households.

Table 6 presents the welfare results from Table 4 and the π gap from Table 5, but after splitting the sample into an early period (1996-2002) and a later period (2002-2018). As noted above, much of the change in home trade shares occurred after 2002 (potentially related to important developments such as China's entry into the WTO in 2001).

Table 6: Welfare Gains and Inflation Gaps

		2002			2018		
Demographic	Characteristic	$\dot{W}_{1996}$	$\dot{W}^{\varphi}_{1996}$	π Gap	$\dot{W}_{2002}$	$\dot{W}^{\varphi}_{2002}$	π Gap
Age	Under Age 30	0.4	0.4	0.4	6.4	7.0	7.2
	Age 30-60	0.2	0.2	0.2	7.5	8.3	9.0
	Over Age 60	1.0	1.1	1.1	5.6	6.1	7.1
Education	H.S. Grad.	-0.1	-0.2	0.0	5.6	6.1	6.9
	College Grad.	-0.2	-0.2	-0.3	7.3	8.1	8.8
	Post-Grad.	1.0	1.0	1.1	6.2	6.9	7.2
Marital Status	Unmarried	-0.2	-0.3	0.0	6.1	6.7	7.3
	Married	0.6	0.7	0.6	6.8	7.4	8.0
Race	White	0.7	0.7	0.7	6.6	7.2	8.0
	Black	-1.8	-1.9	-1.9	5.2	5.7	5.9
	Asian & P.I.	-1.2	-1.4	-1.1	8.0	8.8	9.7
Urban/Rural	Urban	0.3	0.3	0.3	6.4	7.0	7.6
	Rural	1.6	1.8	1.6	7.3	7.9	9.5
	Median	0.3	0.3	0.3	6.4	7.0	7.6

Notes: This table lists welfare gains calculated using Equations (6) and (7) as well as the "π Gap" terms from Table 5 in 2002 (relative to 1996) and in 2018 (relative to 2002).

Interestingly, the results in Table 6 show that in the 1996-2002 period, when home trade shares were relatively stable (and thus taste change is small), $\dot{W}_{1996}, \dot{W}^{\varphi}_{1996}$ and the π Gap between the baseline and the counterfactual are very close. Notably, groups such as Black and Asian households, as well as households headed by someone without a post-grad degree, all show welfare losses from trade and higher overall price inflation from imports in this period. In contrast, other groups had welfare gains from trade.

In the 2002-2018 period, the results in Table 6 show a divergence such that the welfare gains without taste change $\dot{W}_{1996}$ are always smaller than the welfare gains with taste change $\dot{W}^{\varphi}_{1996}$, which are themselves always smaller than the effect on overall price inflation if imports grew the same rate as domestic prices. Still, the qualitative patterns in terms of differences across groups from earlier tables are still evident here.

3.4 Extension: Welfare Analysis for Other Demographics

One potential concern with the results we have presented is that it is possible that there is an underlying additional variable explaining the differences in welfare across groups. Income is a primary concern: it could be that the reason young households have much larger gains from trade than older households is because younger households are more likely to be poorer, and perhaps poorer households benefit more from reductions in trade costs.

We did not use income data from the CE to generate consumer groups based on income, nor did we estimate price indexes for these groups. In order to estimate the contribution of imports to inflation and run our counterfactual experiments fully, it would be necessary to construct the full set of import price indexes by income. However, performing welfare calculations can be done for any demographic characteristic, or mixture of characteristics, given a home trade share and a trade elasticity.

For the purposes of understanding whether income differences are driving the welfare results we presented above, we group households into those with less than $50,000 in income and those with greater than $50,000.[17] We next combine these with age groups to generate six separate consumer groups: Income Below $50,000 and Under 30, Income Below $50,000 and 30-60, Income Below $50,000 and Over 60, and the equivalent 3 groups for households with income above $50,000, and- since income data is not available in 2018- compute home trade shares in 2017 for each of these six subgroups. We again use $\sigma_g = 1.46$ for our welfare calculations. If the welfare differences are primarily caused by income, then we would expect there to be little difference in welfare between age groups within these broad income baskets.

The results are shown in Table 7. First, comparing across ages within income bins does not result in an absence of welfare differences. Relative to autarky, households with a head under 30 still have higher welfare gains than households with a head over 60 in both income bins. That said, the gap in welfare between these age groups (particularly over the observed 1996-2017 period) is larger when conditioning on income above 50,000 dollars (about 6.5 percentage points comparing 1996 to 2017) than it is in the unconditional comparison (about 1.5 percentage points).[18]

[17] There are similar numbers of unique households in each basket when dividing them this way.

[18] Note that potentially due to selection effects, the "average" of the home trade share for households under

Table 7: Home Trade Shares and Welfare Gains from Trade in 2017

	$\lambda_{H,g2017}$	$\dot{W}_A$	$\lambda_{H,g2017} - \lambda_{H,g1996}$	$\dot{W}_{1996}$
Under 30	87.3	29.5	-3.1	7.6
Age 30-60	89.2	24.9	-3.0	7.2
Age Over 60	91.0	20.5	-2.6	6.2
Income Below $50K, Under 30	84.7	36.2	-3.8	9.5
Income Below $50K, 30-60	86.2	32.2	-4.2	10.3
Income Below $50K, Over 60	89.1	25.0	-3.4	8.2
Income Above $50K, Under 30	83.5	39.2	-5.7	14.2
Income Above $50K, 30-60	85.5	33.9	-4.2	10.5
Income Above $50K, Over 60	87.6	28.9	-3.1	7.7

Notes: This table shows home trade shares in 2017, welfare gains from trade in 2017 relative to autarky, the change in home trade shares from 2017 to 1996, and the welfare change from 2017 to 1996 for our 3 age groups, as well as for the 6 subgroups generated by combining age group with whether that household had an annual income above or below $50,000. Source: BLS Consumer Expenditure Survey and authors' calculations.

4 Conclusion

In this paper, we construct "home trade shares" that vary by age, race, marital status, education, and urban status, and use these to analyze differences in inflation and welfare gains from trade for these U.S. demographic groups over the years 1996–2018. We generate these shares by utilizing national import penetration shares for individual imported products, and aggregating these according to group-specific expenditures on those products from the Consumer Expenditure Survey (CE).

We also use the CE and detailed international price data to build import price indexes and overall price indexes that also vary by group. With all these pieces, we can recover trade elasticities that vary by group and perform welfare calculations. We find that imports (inclusive of taste change) held down inflation rates for every group from 1996–2018, but especially so for rural households. Black households benefited the least.

In addition, we estimate that rural households were among those with the largest welfare gains from trade from 1996-2018, while Black households had the lowest gains. For the typical group, more than one quarter of the gains from trade relative to autarky occurred during our time period. Finally, we show that adding taste change to the typical welfare formula boosts the gains from trade for every group relative to the standard welfare results.

30 with income above and below $50,000 is different from the home trade share for households under 30.

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Appendix

A Welfare Derivation

We follow Costinot and Rodríguez-Clare (2014) and define welfare from group g as the value of real consumption: $W_{gt} \equiv \frac{Y_{gt}}{P_{gt}}$. We will be considering "small changes" in these variables to derive an expression for welfare, where a small change is one in which the first-order approximation of a change in a variable captures the effect of the change, meaning for any variable z, $dz = zd\ln z$.

Assuming each group g has some share ω_{gt} of total consumption (such that the shares of all groups in a demographic add to 1), labor market clearing implies $Y_{gt} = \omega_{gt}Y_t = \omega_{gt}w_tL_t$. Normalizing the wage to 1 and assuming that the share ω_{gt} does not change means:

$$d\ln Y_{gt} = d\ln\omega_{gt} + d\ln w_t + d\ln L_t = 0$$

This means that considering changes in prices are sufficient for understanding changes in welfare:

$$d\ln W_t = d\ln Y_{gt} - d\ln P_{gt} = -d\ln P_{gt} \tag{A-1}$$

From Equation (3) above, we have:

$$P_{gt} = \left(\varphi_{H,gt}^{\sigma_g-1}{P_{H,gt}}^{1-\sigma_g} + \varphi_{F,gt}^{\sigma_g-1}{P_{F,gt}}^{1-\sigma_g}\right)^{\frac{1}{1-\sigma_g}}. \tag{A-2}$$

Arkolakis et al. (2012) showed that many of the standard assumptions on the supply side in trade models (Armington, Eaton-Kortum, Melitz) lead to an expression for welfare that only depends on the home trade share and the trade elasticity. Our goal for this section is to demonstrate that this theory still holds even in the case in which there are taste shifters in our price index above, $\varphi_{H,gt}, \varphi_{F,gt}$, so we only present the Armington pricing assumption for this analysis, namely that the underlying price indexes are determined by wages and productivity in Home or Foreign, and some trade cost that can be group-specific:

$$P_{k,gt} = \kappa_{k,gt}\frac{w_k}{A_k}, k = H, F$$

Plugging in to (A-2):

$$\ln P_{gt} = \frac{1}{1-\sigma_g}\ln\left(\sum_{k=H,F}\varphi_{k,gt}^{\sigma_g-1}\kappa_{k,gt}^{1-\sigma_g}w_k^{1-\sigma_g}A_k^{\sigma_g-1}\right)$$

$$d\ln P_{gt} = \frac{\sum_{k=H,F}(1-\sigma_g)\,\varphi_{k,gt}^{\sigma_g-1}\kappa_{k,gt}^{1-\sigma_g}w_k^{1-\sigma_g}A_k^{\sigma_g-1}\times(d\ln\varphi_{k,gt} + d\ln\kappa_{k,gt} + d\ln w_k - d\ln A_k)}{(1-\sigma)\left(\sum_{h=H,F}\varphi_{h,gt}^{\sigma_g-1}\kappa_{h,gt}^{1-\sigma_g}w_h^{1-\sigma_g}A_h^{\sigma_g-1}\right)} \tag{A-3}$$

Note that when introducing taste shifters φ, the first-order approximation assumption made in the above equation is key: for any realized shock between two states, the assumption is that changes in these φ are small enough that the terms can be well approximated by the assumption $d\varphi = \varphi d \ln \varphi$.

Returning to the derivation of welfare, our Armington assumption means that we have a gravity relationship underlying the model. With total purchases for group g from k equal to $Y_{k,gt}$, we have

$$\lambda_{k,gt} = \frac{Y_{k,gt}}{Y_{gt}} = \frac{\varphi_{k,gt}^{\sigma_g - 1} P_{k,gt}^{1-\sigma_g}}{P_{gt}^{1-\sigma_g}} = \frac{\varphi_{k,gt}^{\sigma_g - 1} \kappa_{k,gt}^{1-\sigma} {w_k}^{1-\sigma_g} {A_k}^{\sigma_g - 1}}{\sum_h \varphi_{h,gt}^{\sigma_g - 1} \kappa_{h,gt}^{1-\sigma_g} {w_h}^{1-\sigma_g} {A_h}^{\sigma_g - 1}} \tag{A-4}$$

and we also have:

$$\begin{gathered} \frac{\lambda_{F,gt}}{\lambda_{H,gt}} = \frac{\varphi_{F,gt}^{\sigma_g - 1} \kappa_{F,gt}^{1-\sigma_g} {w_F}^{1-\sigma_g} {A_F}^{\sigma_g - 1}}{\varphi_{H,gt}^{\sigma_g - 1} \kappa_{H,gt}^{1-\sigma_g} {w_H}^{1-\sigma_g} {A_H}^{\sigma_g - 1}} \implies \\ d \ln \lambda_{F,gt} - d \ln \lambda_{H,gt} = (1-\sigma_g)\left(d \ln \varphi_{F,gt} + d \ln \kappa_{F,gt} + d \ln w_F - d \ln A_F\right) \\ - (1-\sigma_g)\left(d \ln \varphi_{H,gt} + d \ln \kappa_{H,gt} + d \ln w_H - d \ln A_H\right) \end{gathered} \tag{A-5}$$

Using (A-4) and (A-5) with (A-3) gives:

$$d \ln P_{gt} = \sum_k \lambda_{k,gt}\left(d \ln \varphi_{k,gt} + d \ln \kappa_{k,gt} + d \ln w_k - d \ln A_k\right)$$

$$= \lambda_F\left(d \ln \varphi_{F,gt} + d \ln \kappa_{F,gt} + d \ln w_F - d \ln A_F\right) + (1-\lambda_F)\left(d \ln \varphi_{H,gt} + d \ln \kappa_{H,gt} + d \ln w_H - d \ln A_H\right)$$

$$= \frac{1}{1-\sigma_g} \lambda_{F,gt}\left(d \ln \lambda_{F,gt} - d \ln \lambda_{H,gt}\right) + \left(d \ln \varphi_{H,gt} + d \ln \kappa_{H,gt} + d \ln w_H - d \ln A_H\right)$$

Since we have assumed that none of the production function parameters in Home are changing over time, only preferences, we have:

$$-d \ln P_{gt} = -\frac{1}{1-\sigma_g}\left(d\lambda_{F,gt} - (1-\lambda_{H,gt})\, d \ln \lambda_{H,gt}\right) - d \ln \varphi_{H,gt}$$

$$= -\frac{1}{1-\sigma_g}\left(-d\lambda_{H,gt} - d \ln \lambda_{H,gt} + d\lambda_{H,gt}\right) - d \ln \varphi_{H,gt} = \frac{1}{1-\sigma_g} d \ln \lambda_{H,gt} - d \ln \varphi_{H,gt}$$

And thus from (A-1) we have the standard Arkolakis et al. (2012) formula without taste changes:

$$d \ln W_{gt} = \ln W_{g1} - \ln W_{g0} = \frac{1}{1-\sigma_g}\left(\ln \lambda_{H,g1} - \ln \lambda_{H,g0}\right)$$

while the version with taste changes is:

$$\ln W_{g1} - \ln W_{g0} = \frac{1}{1-\sigma_g}\left(\ln \lambda_{H,g1} - \ln \lambda_{H,g0}\right) - \left(\ln \varphi_{Hg1} - \ln \varphi_{Hg0}\right)$$

B Expenditure Share Construction

B.1 Domestic Expenditure Shares

We start by replicating publicly available UCC-level expenditures by using the CE microdata for particular characteristic groups. The microdata has households identified by "NEWID", and the data is at the NEWID-UCC expenditure level, both for diary UCCs and for interview UCCs. We have a) a list of UCCs, b) expenditure on some UCCs by NEWID, and c) the characteristics of particular NEWIDs. To build UCC-NEWID expenditures in a way that will lead to proper weighting, for both diary and interview, we:

1. Save a dataset listing the UCCs and expand by the number of NEWIDs, so that the total size of the dataset is # NEWID x # UCCs. (Dataset A)

2. Save a dataset that contains NEWID expenditure on particular UCCs (not the whole set of UCCs). (Dataset B)

3. Make a NEWID-level dataset, keeping the characteristics for each one as well as the weights for each NEWID.

4. Expand the dataset so that each NEWID has a slot for every UCC code.

5. Merge on Dataset A (1:1 _n), so each NEWID has every UCC code.

6. Merge on Dataset B (1:1), so that, where available, each NEWID has expenditure on a particular UCC. Replace missing expenditure observations with zeroes.

Constructing the data in this way means that when applying the calibration weights to each household generates expenditure on each UCC code for any given demographic characteristic.

Next, we decide which characteristics we want to have in the data. We pick characteristics such that the number of household observations underlying each characteristic or set of characteristics is not too small (we try to obtain more than 2,000 NEWID observations in 2018). Table D.1 shows some of the characteristics available in the CE data. We take these variables and convert them into the groupings for the years 1996 through 2018, as shown in Table D.2. Table D.3 shows the household observation counts for each of our chosen groupings in 2018.

Our analysis focuses on 5 demographics in the CE, which are listed in Table D.3 along with the 13 characteristics for the head of household we use and their share within the sample

being used.[19] Household characteristics within a demographic will not overlap. Note that these are not the only characteristics available for these demographics. However, since the PUMD is a survey, the data becomes less representative of the overall population if there are very few respondents.[20] We left out certain characteristics and combined certain other characteristics together for this reason.[21] We additionally prioritized having consistently-defined characteristics over the entire time frame of our analysis.[22] Finally, we note that these characteristics are self-reported: even though about 20% of the U.S. population lives in a rural area according to the official Census Bureau definition, much greater than the share of households in the CE, respondents answer these survey questions according to their own understanding.[23]

B.2 Imported Expenditure Shares

Using the PUMD, we compute UCC product-level expenditures for a host of different characteristic groups for the years 1996 - 2018, and, to line up with the trade data, match those products to HS6 categories using a concordance developed in Furman et al. (2017).[24] We augment the concordance by restricting to only those HS6 categories considered as "consumer goods" under the Broad Economic Categories (BEC) defined by the United Nations.[25] Since these are consumer-facing final goods, the number of HS6 products we obtain is between 440 and 600 depending on the year (out of around 5,000 HS6 products per year in aggregate import data). These matched HS6 products constitute about 31% of total U.S. imports. Finally, in order to estimate a group's import expenditure on a particular HS6 product, we multiply a demographic group's total expenditure in that HS6 in each year by the national

[19] We use the terms "Black/White/Asian households" as shorthand for households headed by a consumer of race Black/White/Asian; of course, other members of the household could be of a different race.

[20] The BLS includes calibration weights in its CE Public Use Microdata for each household in order to be representative of the full CE sample.

[21] We did not include "American Indian" or "Mixed" as characteristics in the Race demographic due to low observation counts. We combined the "Asian" and "Pacific Islander" categories and also selected age ranges. The number of respondents underlying each characteristic we use is summarized in Appendix A, Table D.3.

[22] For example, a variable identifying whether a head-of-household is of Hispanic origin is present in the CE, but was not defined for the entire length of our sample.

[23] See "What is Rural America": `https://www.census.gov/library/stories/2017/08/rural-america.html`.

[24] In cases where multiple HS6 categories map into a single UCC, we distribute expenditures using U.S. import spending on each HS6 as a guide.

[25] See `https://unstats.un.org/unsd/trade/classifications/bec.asp`. The HS6- BEC-Rev. 5 concordance (available here: `https://unstats.un.org/unsd/classifications/Econ/tables/HS2012-17-BEC5_08_Nov_2018.xlsx`) specifies whether the end-use category for any HS6 product is considered a "consumer good".

import penetration rate for that HS6 in each year, where an HS6 import penetration rate is that HS6's import share in domestic absorption (i.e., production minus exports plus imports) at the national level.[26] Annual import and export data at the HS6 level for the United States is freely available from the U.S. Census Bureau, while production data for the United States is available at the NAICS-5 level in the NBER-CES Manufacturing Database from Becker et al. (2013), which we then concord to HS6 codes.[27]

[26]Note that Feenstra and Weinstein (2017) call this denominator both "domestic absorption" and "apparent consumption".

[27]The HS6 category for gasoline/petroleum products changes multiple times over our time period. Due to its importance for consumer baskets, we ensure that this product remains in our sample by calculating the import penetration ratio for each category in each year, and then harmonize them into a single HS6 code for all years for concording to CE data.

C Import Price Index Construction

C.1 Import Price Indexes under Nested CES Preferences

In order to construct import price indexes, we use U.S. import data on prices, sales, and expenditures with a nested Constant Elasticity of Substitution demand system with group-level taste shifters at the HS6 sectoral level. The structure of the model is similar to Hottman and Monarch (2020). However, there are several important differences. One difference from that prior work (which was focused on income group differences using annual price data) is that we build our price indexes and estimate the parameters of our model using *quarterly* U.S. Census data on the universe of prices and sales of foreign suppliers exporting individual HS10 products to the United States. We make a number of improvements to the data and methodology relative to Hottman and Monarch (2020): we use annual group-level expenditure shares from the Consumer Expenditure Survey Public Use Micro-Data for each year instead of just the 2014 group-level expenditure shares, estimate the model for about 650 HS6 imported products instead of 250 HS4 sectors, adjust the concordance of Furman et al. (2017) to include only consumer goods under the Broad Economic Categories defined by the United Nations, and use quarterly data on prices instead of annual data.

We assume U.S. households have ordinary CES preferences over imported sectors, such that the utility of group g at time t is given by

$$V_{gt} = \left[\sum_{s \in \overline{S}_t} \varphi_{gst}^{\frac{\sigma_F - 1}{\sigma_F}} Q_{gst}^{\frac{\sigma_F - 1}{\sigma_F}} \right]^{\frac{\sigma_F}{\sigma_F - 1}} \tag{C-1}$$

where V_{gt} is the CES aggregate of real consumption of tradable consumer goods sectors for group g at time t, Q_{gst} is the consumption index of sector s for household group g at time t, $\varphi_{gst} > 0$ is a taste parameter for sector s for group g at time t, $\overline{S}_t$ is the set of available sectors, and $\sigma_F > 0$ is an elasticity parameter. At this tier, differences across groups in spending patterns is generated from the sector-specific taste shifters at the group level, φ_{gst}.

The consumption index of sector s for group g at time t is

$$Q_{gst} = \left[\sum_{v \in \overline{G}_{s,t}} \varphi_{vt}^{\frac{\sigma^s - 1}{\sigma^s}} q_{gvt}^{\frac{\sigma^s - 1}{\sigma^s}} \right]^{\frac{\sigma^s}{\sigma^s - 1}} \tag{C-2}$$

where q_{gvt} is the real consumption of variety v in sector s for group g at time t, $\varphi_{vt} > 0$ is a demand shifter for variety v at time t, $\overline{G}_{s,t}$ is the set of available varieties in sector s, and $\sigma^s > 0$ is an elasticity parameter for sector s.

The utility maximizing quantity demanded of variety v in sector s for group g at time t is

$$q_{gvt} = \left(\frac{{\varphi_{vt}^{s}}^{\sigma^s - 1} {p_{vt}}^{-\sigma^s}}{P_{st}^{1-\sigma^s}} \right) Y_{gst}, \tag{C-3}$$

where Y_{gst} is the expenditure on imported sector s for group g at time t, and p_{vt} is the variety-specific price at time t, and P_{st} is a sectoral price aggregate over varieties given by

$$P_{st} = \left(\sum_{v \in \overline{G}_{s,t}} \varphi_{vt}^{\sigma^s - 1} p_{vt}^{1-\sigma^s} \right)^{\frac{1}{1-\sigma^s}}. \tag{C-4}$$

The utility maximizing expenditures of group g on sector s is:

$$Y_{gst} = \left(\frac{{\varphi_{gst}^{s}}^{\sigma_F - 1} {P_{st}}^{1-\sigma_F}}{P_{gt}^{1-\sigma}} \right) Y_{gt}, \tag{C-5}$$

where Y_{gt} is the total expenditure of group g at time t and P_{gt} is the group-level price aggregate. We allow the sector-level taste shifters (φ_{gst}) to be different across groups.

We have the following price aggregate for group g's imported consumption:

$$P_{F,gt} = \left(\sum_{s \in \overline{S}_t} {\varphi_{gst}^{s}}^{\sigma_F - 1} {P_{st}}^{1-\sigma_F} \right)^{\frac{1}{1-\sigma_F}} \tag{C-6}$$

As a technical matter, we form chained import price indexes first for any sector s by normalizing the first quarter of 1998 to a value of 100, and then taking cumulative products of the chain links, where the link for time t is given by

$$\frac{P_{st}}{P_{st-1}} = \frac{\left(\sum_{j \in \overline{G}_{s,t}} p_{jt}^{1-\sigma^s} {\varphi_{jt}}^{\sigma^s - 1} \right)^{\frac{1}{1-\sigma^s}}}{\left(\sum_{j \in \overline{G}_{s,t}} p_{jt-1}^{1-\sigma^s} {\varphi_{jt-1}}^{\sigma^s - 1} \right)^{\frac{1}{1-\sigma^s}}}. \tag{C-7}$$

Since we are using quarterly data at the supplier-product level, we reduce measurement error by: 1) dropping observations of supplier-product price changes that are extreme and 2) restricting our price index only to those supplier-product combinations that are available in a set of consecutive quarters[28]. This ensures that small varieties that pass in and out from the data do not lead to outsize effects in the quarters they enter and exit, while still allowing for the the entry of new products into the consumption index over time. Thus we define

[28] We could make other choices to address this problem, such as continuing to use the last available price for products that enter and exit the data frequently.

$G_{s,t}$ to be the set of varieties in sector s at time t and take $\overline{G}_{st} = \{G_{s,t+1} \cap G_{s,t} \cap G_{s,t-1}\} \cup \{G_{s,t} \cap G_{s,t-1} \cap G_{s,t-2}\}$ to be the set of varieties at time t that are a) also present in $t-1$ and b) present for at least 3 consecutive quarters (where t and $t-1$ are 2 of those quarters). Finally, in order to seasonally adjust them, we also remove sector-specific quarterly fixed effects from the sector price indexes before we form the group-level price indexes.

As we did for sector s, we make the same normalization and use the same chaining procedure to form the group level import price indexes, with the time t link given by

$$\pi_{F,gt} = \frac{P_{F,gt}}{P_{F,gt-1}} = \frac{\left(\sum_{s \in \overline{S_t}} \varphi_{gst}^{\sigma-1} {P_{st}}^{1-\sigma}\right)^{\frac{1}{1-\sigma}}}{\left(\sum_{s \in \overline{S_t}} \varphi_{gst-1}^{\sigma-1} {P_{st-1}}^{1-\sigma}\right)^{\frac{1}{1-\sigma}}}. \tag{C-8}$$

We also define the set of tradable consumer goods sectors at time t to be S_t, with $\overline{S}_t = \{S_{t+1} \cap S_t \cap S_{t-1}\} \cup \{S_t \cap S_{t-1} \cap S_{t-2}\}$ the set of sectors at time t that are a) also present in $t-1$ and b) present for 3 consecutive quarters.

To construct the CES import price index for each group g in each period ($P_{F,gt}$) as in Equation (C-8), we need a measure of the taste shifter φ_{gst}, an estimate of σ_F, and the sector level price indexes P_{st}, which themselves require the variety-level taste shifter φ_{vt} and the sector-level elasticity of substitution σ^s.

In order to estimate these parameters, we close the model by assuming a basic monopolistic competition setting with decreasing returns to scale that gives the producer of variety v the following pricing equation:

$$p_{vt} = \frac{\sigma^s}{\sigma^s - 1} \delta_{vt} (1 + \omega^s) q_{vt}^{\omega^s} \tag{C-9}$$

We can then apply the Feenstra (1994) estimation strategy, discussed next, to recover the key parameters of the model.

C.2 Estimation

In this section, we describe how we recover σ^s, ω^s, and φ_{vt}, addressing endogeneity concerns by applying the Feenstra (1994) approach of identification via heteroskedasticity to estimate the model.

For each sector s, the deep parameters to be estimated are σ^s and ω^s. Conditional on estimating these parameters, the variety-level unobservables of φ_{vt} (demand shifters) can be recovered from the model's structure given data on prices and sales.

Start from the variety-level demand expression in Equation C-3. Taking logs, taking the time difference and differencing relative to another variety k in the same sector s gives

$$\Delta^{k,t} \ln(p_{vt} q_{vt}) = (1 - \sigma^s) \Delta^{k,t} \ln(p_{vt}) + \nu_{vt}, \tag{C-10}$$

where $\Delta^{k,t}$ refers to the double difference. The unobserved error term is $\nu_{vt} = (1 - \sigma^s) [\triangle^t \ln \varphi_{kt} - \triangle^t \ln \varphi_{vt}]$, where $\triangle^t$ refers to a single difference across time periods.

Next, we work with the variety-level pricing expression in Equation C-9. Multiplying both sides by $p_{vt}^{\omega_s}$, taking logs, re-arranging, and double-differencing as before gives

$$\Delta^{k,t} \ln p_{vt} = \frac{\omega^s}{1 + \omega^s} \Delta^{k,t} \ln(p_{vt} q_{vt}) + \kappa_{vt}, \tag{C-11}$$

where the unobserved error term is $\kappa_{vt} = \frac{1}{1+\omega^s} [\triangle^t \ln \delta_{vt} - \triangle^t \ln \delta_{kt}]$.

As in Feenstra (1994), we assume that the following orthogonality condition holds for each variety:

$$G(\beta_s) = \mathbb{E}_{\mathbb{T}} [x_{vt}(\beta_s)] = 0 \tag{C-12}$$

where $\mathbb{E}_{\mathbb{T}}$ is the time series expectation, $x_{vt} = \nu_{vt} \kappa_{vt}$, and $\beta_s = \begin{pmatrix} \sigma^s \\ \omega^s \end{pmatrix}$.

In words, we are assuming the orthogonality of the idiosyncratic demand (ν_{vt}) and supply (κ_{vt}) shocks at the variety level, after variety and sector-time fixed effects have been differenced out. The supply shock κ_{vt} is the residual of the pricing equation after accounting for fixed effects and the variation in prices due to movements along upward-sloping supply curves. The supply shock κ_{vt} thus represents shifts over time in the intercept of the variety-level supply curve- changes in price over time that occur for reasons other than changes in quantity sold. Our assumption is that these intercept shifts are uncorrelated with shifts over time in the intercept of the variety-level demand curve. This orthogonality assumption is plausible in our setting using supplier-product trade data.

The objective function is formed for each sector s by stacking the orthogonality conditions, so that the GMM problem is:

$$\hat{\beta}_s = \arg \min_{\beta_s} \{G^*(\beta_s)' W G^*(\beta_s)\} \tag{C-13}$$

where $G^*(\beta_s)$ is the sample counterpart of $G(\beta_s)$ stacked over all varieties in sector s and W is a positive definite weighting matrix[29]. Following Broda and Weinstein (2006), we give more weight to varieties that are present in the data for longer time periods and sell larger quantities[30].

[29] In principle, one could use the optimal GMM weighting matrix, but optimal GMM is known to have a serious small-sample bias in a setting like ours.

[30] Varieties with larger import volumes are expected to have less measurement error in their unit values.

After obtaining ω^s and σ^s for each sector from the GMM estimation, we can recover the variety-level demand shifters. Although most papers in the literature on price index construction impose the assumption that variety-level quality is fixed over time, Redding and Weinstein (2020) show that the price index is still well-behaved so long as variety-level quality is unchanged on average for the common set of varieties. Therefore, in this spirit, we normalize the geometric average of demand shifters across varieties in each sector ($k \in \overline{G}_{s,t}$) to be 1 (i.e., $\widetilde{\varphi_{kt}} = 1$ for all time periods, where the tilde denotes the geometric average). Then, the demand shifter for each variety can be computed differencing Equation C-3 relative to the geometric average to get the following expression

$$\varphi_{vt} = \exp\left[\frac{\ln(p_{vt}q_{vt}) - \ln(\widetilde{p_{kt}q_{kt}}) + (\sigma^s - 1)(\ln p_{vt} - \ln \widetilde{p_{kt}})}{\sigma^s - 1}\right], \tag{C-14}$$

where $(\widetilde{p_{kt}q_{kt}})$ is the geometric average of $(p_{kt}q_{kt})$ across varieties in the sector ($k \in \overline{G}_{s,t}$) at time t, and similarly for $(\widetilde{p_{kt}})$. Importantly, the variety-specific taste parameters scale in units of prices, thereby allowing comparison of prices across varieties.

With the previously estimated parameters as well as constructed expenditure on imports by sector, Y_{gst}, it is possible to estimate the overall elasticity of substitution σ_F. Starting from the group sector-level demand expression in Equation C-5, take the time difference and difference relative to another sector k bought by the same group h. This double-differencing gives

$$\Delta^{k,t} \ln(Y_{gst}) = (1 - \sigma_F)\Delta^{k,t} \ln(P_{st}) + \upsilon_{gst}, \tag{C-15}$$

where $\upsilon_{gst} = (\sigma_F - 1)\left[\Delta^{k,t} \ln \varphi_{gst}\right]$. We can construct the objects that enter this equation using our estimated parameters and the data. We then form our estimating equation by pooling the double-differenced observations across groups, sectors, and time. Note also that this equation depends only on relative log-changes, such that time-invariant differences across sectors do not affect this equation.

We expect that running Ordinary Least Squares on the above equation would not produce a consistent estimate of σ, because of potential endogeneity bias from a possible correlation between the sectoral price index and the error term. To address this potential issue, we pursue an instrumental variables approach as in Hottman et al. (2016). Note that the change in the log of the sectoral price index can be linearly decomposed into two terms as follows:

$$\Delta^{k,t} \ln P_{st} = \Delta^{k,t} \left(\frac{1}{N_{st}^v} \sum_{v \in \overline{G}_{s,t}} \ln p_{vt} \right) - \Delta^{k,t} \frac{1}{\sigma^S - 1} \ln \left(\frac{1}{N_{st}^v} \sum_{v \in \overline{G}_{s,t}} \frac{(\frac{p_{vt}}{\varphi_{vt}})^{1-\sigma^S}}{\widetilde{(\frac{p_{vt}}{\varphi_{vt}})^{1-\sigma^S}}} \right), \tag{C-16}$$

where N_{st}^v is the number of varieties in $\overline{G}_{s,t}$. We use the second term on the right-hand side, which measures the change in dispersion in quality-adjusted variety-level prices within a sector, as an instrument for the change in the price index term when we estimate Equation C-15. We use this term because the first term on the right-hand side is likely correlated with changes in the sector-level demand shifter, as average prices rise in response to positive sector demand shocks. Our identifying assumption for the instrumental variables regression is that the changes in dispersion in quality-adjusted prices within a sector are uncorrelated with the changes in the sector-level demand shifter φ_{gst}. Based on that assumption, we estimate Equation C-15 using two-stage least squares to obtain an estimate of σ.

Given an estimate of σ_F, we can then finally generate the group-specific sectoral demand shifters (φ_{gst}) using our data on group-level expenditure described in Section 2.4.1. Along the lines of the above calculation of φ_{vt}, this can be done by normalizing the geometric average of demand shifters across sectors ($s \in \overline{S_t}$) to be 1 (i.e., $\widetilde{\varphi_{gkt}} = 1$ for all time periods) and using Equation C-5 in differences to derive

$$\varphi_{gst} = \exp \left[\frac{\ln(Y_{gst}) - \ln(\widetilde{Y_{gkt}}) + (\sigma_F - 1)(\ln P_{st} - \ln \widetilde{P_{kt}})}{(\sigma_F - 1)} \right] \tag{C-17}$$

Therefore, the sectoral level demand shifters are a function of group expenditure on imports in that sector (Y_{gst}) and the accompanying sectoral price index (P_{st}), relative to that group's geometric average across sectors ($s \in \overline{S_t}$). These taste parameters φ_{gst} drive the differences in import prices between groups.

The estimated elasticities of substitution (σ^s) are key parameters in the nested CES price indexes. We find a mean elasticity of 4.2, with a standard deviation of 2.8. These estimates are quantitatively similar to those reported in Hottman and Monarch (2020), and that paper further shows that its estimates are comparable to benchmark estimates from the literature.

Table D.4 reports estimates of the aggregate elasticity of substitution across imported sectors. The OLS estimate of this parameter is 1.1, while our IV approach yields a point estimate centered at 1.78 with a 95 percent confidence interval between 1.75 and 1.81. These estimates are quite similar to those in Hottman and Monarch (2020), although slightly higher than the elasticity of 1.36 reported by Redding and Weinstein (2024). Given this elasticity of substitution across sectors, we can solve for group-specific sectoral demand shifters (φ_{gst}) as in equation C-17, allowing the computation of import price indexes across groups.

D Appendix Tables

Table D.1: List of Characteristics, CE data

Demographic	Variable	Value	Availability
Age	AGE	Numeric Age	1990-
Education	EDUC_REF	12= High School Graduate 13 = Some College, no degree 14 = Associate's degree 15 = Bachelor's degree 16 = Master's, Professional, or Doctorate degree	1996-
Marital Status	MARITAL	1 = Married 2 = Widowed 3= Divorced 4=Separated 5=Never Married	1996-
Race	RACE / MEMRACE	1=White 2=Black 3=Amer. Ind. 4=Asian	1990 -
Urban	BLS_URBN	1=Urban, 2=Rural	1984-

Table D.2: List of Characteristics for Import Data

Demographic	Characteristic	Number of Groups
Age Range	1= <30 , 2=30-60 , 3=>60	3
Education	2= H.S. Graduate 3= College Graduate 4= Post-Graduate	4
Marital Status	1 = Unmarried 2=Married	2
Race	1=White 2=Black 3=Asian and P.I.	3
Urban	1=Urban 2=Rural	2

Table D.3: Selected Demographic Characteristics in the Consumer Expenditure Survey

Demographic	Characteristic	Number of Households (2018)
Age	Under Age 30	4,565
	Age 30-60	21,228
	Over Age 60	13,918
Education	High School Graduate	17,246
	College Graduate	13,183
	Post-Graduate	5,429
Marital Status	Unmarried	19,063
	Married	20,648
Race	White	32,506
	Black	4,173
	Asian & P.I.	2,205
Urban/Rural	Urban	37,189
	Rural	2,672

Notes: This table summarizes the number of households for a particular generated characteristic within the selected demographic group.

Table D.4: Summary of σ_F

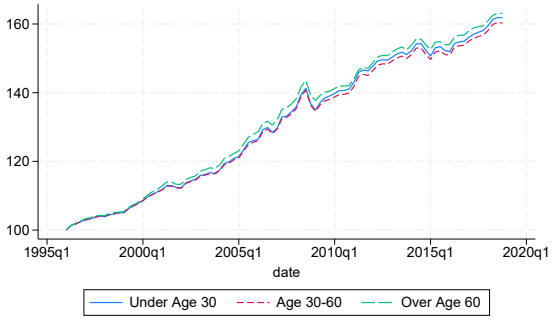
OLS	IV	95% Confidence Interval
1.09	1.78	$[1.75, 1.81]$

Source: LFTTD and authors' calculations.

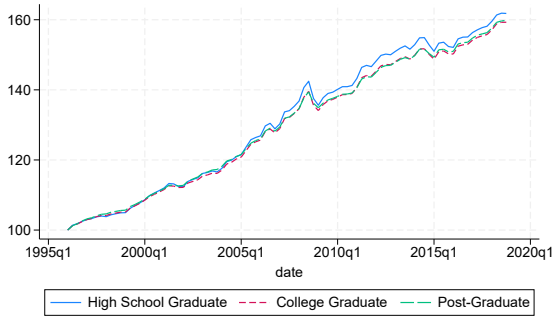
E Appendix Figures

Figure E.1: Overall Price Index

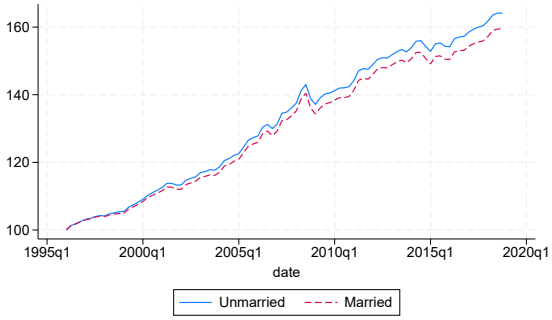
(a) Age



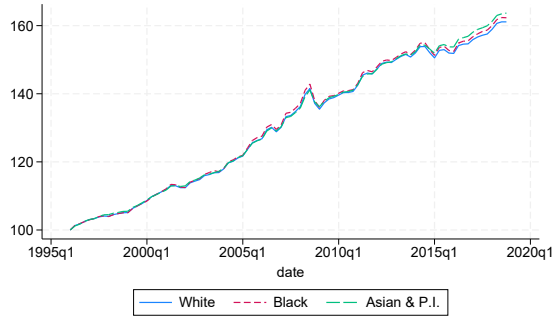
(b) Education



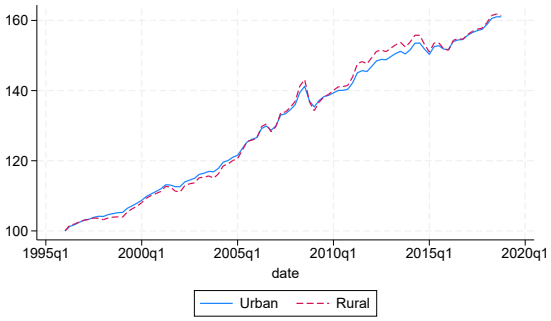
(c) Marital Status



(d) Race



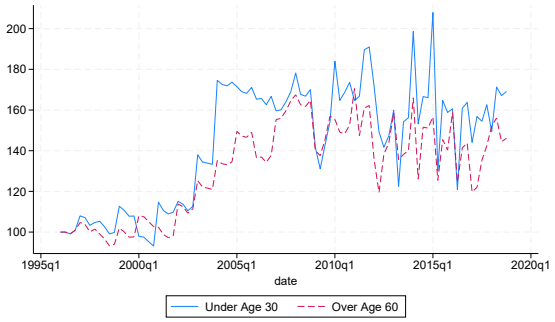
(e) Urban/Rural



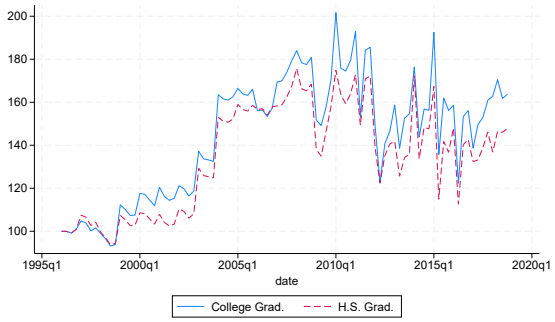
Notes: This figure plots the overall price index for different demographic characteristics. Source: BLS CPI and Consumer Expenditure Survey and authors' calculations.

Figure E.2: Import Price Indexes by Group

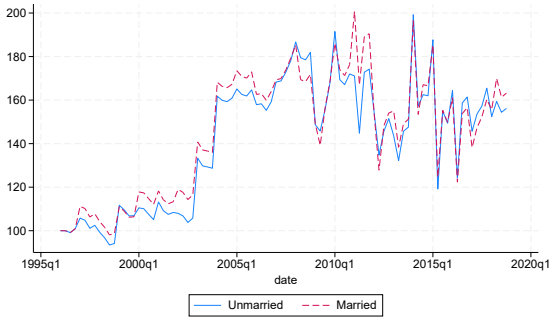
(a) Age



(b) Education



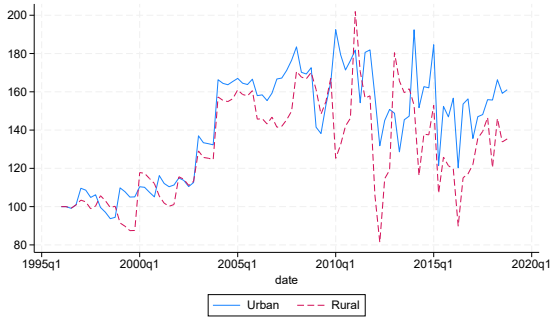
(c) Marital Status



(d) Race

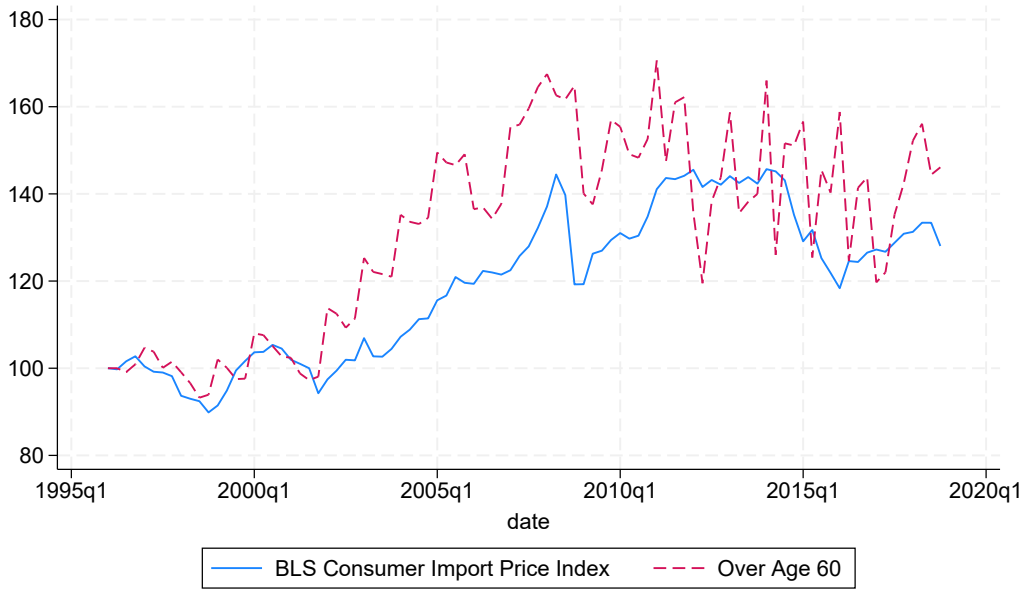


(e) Urban/Rural



Notes: This figure plots the estimated quarterly import price indexes from 1996 Q1 through 2018 Q4 for selected characteristics within a given demographic category. The price indexes are indexed to 100 in 1996 Q1. Source: LFTTD, BLS Consumer Expenditure Survey and authors' calculations.

Figure E.3: BLS Consumer Import Price Index



Notes: This figure plots a BLS Consumer Import Price Index for from 1996 to 2018 against one of our estimated group-level import price indexes ("Over Age 60"). We construct the BLS Consumer Import Price Index as a national, consumer product version of BLS import prices generated by equally weighting the end-use categories "Foods, Feeds, and Beverages", "Fuels and Lubricants", and "New and used cars", and then equally weighting that block with the end-use category "Consumer goods ex Autos" (which is roughly in-line with 2019 trade shares). Data is available at `https://www.bls.gov/web/ximpim/beaimp.htm`. Source: LFTTD, BLS Import Price Indexes and authors' calculations.